



United International University

Department of Computer Science and Engineering

CSE 2213/CSI 219: Discrete Mathematics (BSCSE and BSDS)

Mid-term Examination – Fall 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. (a) Construct a truth table for the following expression: $(p \vee \neg q) \rightarrow q$ [2]
(b) State the contrapositive and the inverse of the following conditional statement. [2]
“I come to class when there is going to be a quiz.”
(c) Show that $\neg p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$ using logical equivalence laws. [2]
2. (a) Consider the following predicates: [1*4=4]

$T(x)$: x is a teacher
 $L(x)$: x loves teaching
 $C(x)$: x has a classroom
 $R(x)$: x is respected

The domain consists of all people. Express the following statements using quantifiers, logical connectives, and the predicates above:

- (i) All teachers in the city love teaching.
(ii) Anyone who is respected must be a teacher or love teaching.
(iii) Not everyone who loves teaching is a teacher.
(iv) Some teachers who have classrooms does not have respect.
- (b) Determine whether the following propositions are true or false. Justify your answer. The domain of all variables is the set of real numbers. [1*2=2]
- (i) $\forall x \exists y \forall z (x^2 + y^2 \geq z)$
(ii) $(\forall x (x^2 \leq x^2 + 1)) \leftrightarrow (\exists y (y^2 > y^2 + 1))$
3. Let's assume: $A = \{x \in \mathbb{Z}^+ | x^2 < 10\}$, $B = \{1, 3, 5, 7, 9\}$, $C = \{3, 4, 5, 9\}$
- (a) Identify the elements in set A. [1]
(b) Find the power set of $(A \cap C)$. [1]
(c) Find the set $(A \cap B) \cup (B \cap C)$. [2]
(d) Find the set $(A - B) \times (C - A)$. [2]
4. (a) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x}$. Is this a function? Justify your answer. [2]
(b) Show that the following function is invertible. Find its inverse function $f^{-1}(x)$, and evaluate $f^{-1}(2)$. [4]

$$f : \mathbb{R} - \{-2\} \rightarrow \mathbb{R} - \{3\}, \quad f(x) = \frac{3x - 4}{x + 2}.$$

5. (a) Prove the following statement using an appropriate proof technique: “If a and b are two odd integers, then the sum of the squares of a and b is even.” [3]
(b) Consider the proposition, $P(n)$: If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$. Prove that $P(1)$ is true. What kind of proof did you use? [3]